A Hybrid Siamese Network With Spatiotemporal Enhancement and Two-Level Feature Fusion for Remote Sensing Image Change Detection

Liangliang Yan and Jie Jiang

***Abstract*— With the popularization and development of deep learning (DL) technology, remote sensing (RS) image change detection (CD) has achieved remarkable success. However, an accurate CD has still been challenging due to the difficulties in achieving efficient feature extraction and effective difference feature enhancement and refinement. To address these limitations, this article proposes a hybrid Siamese network with spatiotemporal enhancement and two-level feature fusion (named the HSSENet) for CD. First, an efficient hybrid Siamese backbone is designed by combining a transformer's advantage to capture dense dependencies between features and convolutional neural network (CNN)'s advantage to provide local prior knowledge. In addition, to reduce irrelevant pseudo-changes and high-frequency noise while maintaining the high compactness of changed targets, a spatiotemporal enhancement module (STEM) that adopts the self-attention mechanism for context modeling in spatiotemporal dimensions and can separately process low and high frequencies is proposed for effective difference feature enhancement. Finally, three two-level feature fusion modules (TL-FFMs) are designed instead of standard decoders to aggregate low-level details and high-level semantics for refining the boundary information. The proposed HSSENet is verified by experiments, and the experimental results demonstrate that it can obtain a better tradeoff between accuracy and efficiency than the state-of-the-art methods and significantly outperforms them with the F1-score of 91.48/91.55/91.17 points on the learning, vision, and RS (LEVIR)/Wuhan University (WHU)/deeply supervised image fusion network (DSIFN) test sets, respectively.**

***Index Terms*— Change detection (CD), hybrid Siamese backbone, spatiotemporal enhancement module (STEM), two-level feature fusion module (TL-FFM).**

I. Introduction

REMOTE sensing (RS) image change detection (CD) has been one of the popular research directions in RS image analysis. The CD aims to identify differences between two co-registered and co-corrected RS images acquired in the same scene [1] and determine where and which type of changes have occurred. The former belongs to the field of binary CD, which is also the research objective of this study. Its purpose is to obtain a binary CD map, so it represents a pixel-level classification task. The latter belongs to the semantic CD [2], [3], [4], [5]. Its purpose is to obtain a multiclass CD map, so it is more like a combination of the semantic segmentation and binary CD. With the development of data acquisition platforms, such as satellites and ground survey vehicles, their massive multisource RS data have brought new application requirements into the field of land change monitoring [1]. In particular, multisensor high-resolution (HR) data require highly automated and powerful CD methods to reduce the cost of manual interpretation. In recent years, the CD has gradually become the main research method of global surface change. It has been widely used in many fields, including urban management [6], [7], urban expansion estimation [8], deforestation [9], environmental monitoring [10], [11], disaster prevention and mitigation [12], [13], damage assessment [14], [15], and others.

The existing CD methods can be roughly classified into traditional methods and deep learning (DL)-based methods. Before the advent of DL technology, traditional image processing technology was generally adopted. Due to the low resolution of early RS images, researchers often implemented the CD task in the pixel space. The difference images were obtained by performing some algebraic operations (e.g., image difference and image ratio) on RS data. Then, threshold-based [16] or clustering-based (k-means [17]) image segmentation techniques were used to obtain binary CD maps. Alternatively, the transformation-based methods, including the principal component analysis (PCA) [17], [18], change vector analysis (CVA) [19], and multivariate alteration detection (MAD) [20], [21], have been introduced to the field of CD to enhance the changed information of bitemporal images and suppress irrelevant information. With the development of high-precision sensors, the acquisition of HR RS images has been increasingly accessible, and the methods based on machine learning (ML) have attracted great research attention. These methods, including the Bayesian classification [22], support vector machines (SVMs) [23], decision trees [24], random forest [25], and k-nearest neighbor (KNN) [26], have been increasingly employed to improve the expression and recognition of features. In addition, the probabilistic graphical models, such as conditional random field (CRF) [27] and Markov random field (MRF) [28], have been proven to be effective

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in CD task. Although the ML-based CD methods improve the detection accuracy to a certain extent, complex feature engineering algorithms are still needed, and the generalization performance is poor.

Over the past few years, DL has achieved significant success in the field of computer vision (CV). In addition, the DL-based techniques, represented by convolutional neural networks (CNNs) [29], generative adversarial networks (GANs) [30], [31], and recurrent neural networks (RNNs) [32], [33], have been gradually introduced into the CD field. The DL-based CD methods do not require manual feature design and rely on the powerful modeling and learning capabilities of neural networks to simulate the relationship between image objects and relevant real-world geographical features. They learn high-level abstract semantic information from massive RS data to achieve end-to-end change information recognition. Due to the efficient feature extraction capability of CNNs, they have been employed in most recent supervised CD methods to extract deep semantic features of bitemporal images separately to identify changes in data of interest [34]. Inspired by semantic segmentation models [35], [36], the encoder–decoder structure-based methods have been widely used in RS image interpretation [37], [38]. Similar to the semantic segmentation task, improving the contextual representations of models is extremely vital for the recognition of changed targets. Therefore, the latest studies have been gradually devoted toward enlarging the receptive field (RF) size [39] and using the attention mechanism (spatial or channel attention [40], [41], [42], [43]) to weigh the feature maps. These methods can improve the detection accuracy to a certain extent. However, due to the essential properties of local RF in CNNs, they still rely on stacking network layers to enlarge the RF size, which increases the number of model parameters and computational overhead significantly. Very recently, an emerging model, transformer (TR), has shown extraordinary prospects in natural language processing (NLP) and CV fields. A number of cutting-edge scholars have innovatively introduced TR into the CD field [44], [45], [46]. Unlike the CNN-based methods, the TR-based methods use a self-attention mechanism to establish long-distance relationships between features during feature extraction, showing a more powerful contextual representation capability compared to the CNNs.

Based on the above analysis, it can be concluded that many DL-based methods have been applied to the RS image CD, achieving great breakthroughs, but there have still been certain open issues, and thus, further improvement is needed. One of the problems of the existing CD methods is an inefficient and inadequate feature representation. Namely, using a standard CNN structure as a backbone struggles to model dependencies between long-range features, which can easily lead to the ambiguity and omission of predicted changes. In addition, applying an original TR architecture for feature extraction has the disadvantages of numerous model parameters and high computational overhead. A straightforward idea for solving these challenges could be to combine CNN and TR. Recent research has combined these two structures, connecting them in series [46], [47], but the proposed solutions are still suboptimal because they lack sufficient multilevel integration of the two models. Further, cutting-edge researchers have tried to use the TR to model the spatial–temporal correlation of bitemporal features aiming to enhance difference feature representations [46], [48], [49]. However, they mainly used the TR blocks directly to achieve the same global attention on each token without considering the unique characteristics of image space that are different from those of the text space. For instance, image features contain different base frequencies, which easily results in high-frequency burrs and low compactness in the predicted changed objects. Finally, in regular decoders, either original low-level information is easily overlooked, or the fusion effect of multilevel features is not efficient. The former may cause the uncertainty of boundary pixels of changed targets, while the latter may cause a certain amount of waste of computing resources.

To overcome the aforementioned problems, this study proposes a hybrid Siamese network with spatiotemporal enhancement and two-level feature fusion (HSSENet) for RS image CD. The main contributions of this work are as follows.

1) An efficient Siamese backbone for the CD task is designed using a lightweight CNN branch to provide local prior knowledge and a TR branch to capture long-range dependencies. This Siamese backbone can help to obtain discriminative object-level semantic features and achieve a balance between performance and overhead.
2) Based on the self-attention mechanism, the spatiotemporal enhancement module (STEM) is adopted to model the spatiotemporal relationship between bitemporal features and enhance original difference features in high–low (HiLo) frequencies, which benefits in maintaining a compact structure of changed objects and suppressing high-frequency noise and pseudo-changes.
3) Unlike regular decoding methods, the proposed two-level feature fusion modules (TL-FFMs) use high-level context-rich features to guide the two-level feature fusion and recover spatial edge details in the decoding stage successively, which is essential for the refinement of target edges.

The rest of this article is organized as follows. Section II reviews recent work on the CNN- and TR-based methods in the RS image CD. Section III describes the proposed HSSENet in detail. Section IV presents the experimental results, discussion of pivotal issues, and network visualization analysis. Finally, Section V concludes this article.

II. Related Work

There have been a variety of classification strategies for DL-based CD methods. According to whether they require training samples with labels for learning, these methods can be divided into unsupervised CD methods [31], [50] and supervised CD methods [37], [51]. In addition, some studies have classified CD models according to the model structure into single-stream models [52], [53], [54], which use fused bitemporal images as input, and double-stream models [55]. The latter generally consists of two subfeature extractors with the same structure, named a Siamese backbone, which

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converts bitemporal images into feature maps. Then, contrastive analysis is performed to generate the CD map. According to whether the weights of two subfeature extractors are shared [56], the double-stream models can be further divided into those with a Siamese structure and those with a pseudo-Siamese structure. Aiming to alleviate model overhead, the pure Siamese-structure-based methods have been favored in recent studies. Considering the advent of the emerging TR, this article roughly divides the recent CD methods into the following two categories from a new perspective, namely, CNN- and TR-based CD methods.

A. CNN-Based CD Methods

Due to the unique visual inductive bias and efficient feature representation capability of the CNNs, CNN-based approaches have been widely proposed for RS image CD. For instance, Daudt et al. [37] presented three fully convolutional neural networks (FCNs), which denoted a big step toward HR RS image interpreting. Subsequently, since context modeling is crucial for the identification of changed objects of interest, Song et al. [52] adopted dilated convolution and deformable convolution instead of standard convolution to enlarge the RF size. Furthermore, the latest research has been more enthusiastic about designing various component-level attention modules [40], [41], [42], [43] to reweight the attention of the feature maps in the spatial or channel dimensions and has ulteriorly given more attention to task-related locations and channels. Multilevel feature fusion is also important for multiscale detection. To leverage the shallow-layer fine-grained information, Fang et al. [41] designed a dense information transmission structure to reduce the uncertainty of edge pixels in changed targets. Alternatively, a number of studies have applied unique loss functions to solve the problem of sample imbalance. Liu et al. [42] integrated the semantic segmentation task and the CD task, and they adopted an improved focal loss function to redistribute the weights of changed samples and unchanged samples.

Although the CNN-based methods have improved the feature representation capability of models to a certain extent, they are mostly based on the architectural system of FCNs, making them limited by the intrinsic local RF characteristics of the CNNs. Therefore, they struggle to model the dense long-distance relationships between image features. In addition, integrating complex attention modules at each stage of CD models or implementing dense information transmission also increases colossal computational overhead.

B. TR-Based CD Methods

The TR, proposed by Vaswani et al. [57] in 2017, was originally widely used in the field of NLP for sequence prediction and has been gradually introduced into CV tasks in recent years. Unlike regular attention mechanisms, the TR employs a nonlocal attention mechanism, termed self-attention. In view of the internal correlation between features, the TR can model the dependencies of any positions of feature vectors or maps, so it is convenient for long-range contextual relationship modeling.

After witnessing the astounding performance of the emerging TR in various CV tasks, some cutting-edge scholars have been gradually discussing the application of the TR to the RS image CD. Recent TR-based CD methods can be roughly divided into two categories, of which one completely abandons the model architecture of deep CNNs and adopts the pure TR network for CD. For instance, Bandara and Patel [45] used hierarchical structured TR as an encoder and multilayer perception (MLP) as a decoder. Zhang et al. [58] used the Swin TR blocks as a basic unit of the encoder and decoder. These methods can significantly improve the detection performance but have high computational complexity. Another category first uses CNNs to perform feature extraction and then employs the TR to enhance feature representations. Chen and Shi [44] performed a spatial–temporal attention in each subregion of bitemporal images to obtain more discriminative features. They also used a modified lightweight ResNet 18 as the backbone, and the last-level outputs of the Siamese backbone were fed to standard TR encoder–decoder blocks to enhance difference features in the token-based space [46]. However, because of inadequate feature extraction and the neglection of low-level feature maps, its performance is not satisfactory and can be further improved. Wang et al. [48] adopted the ResNet 50 for feature extraction, and the output of each stage of the ResNet 50 was fed to TR blocks for separate enhancement in spatial and temporal dimensions, which can effectively suppress the irrelevant change information in complex scenes, but relies heavily on the large-scale and computation-consuming ResNet 50 backbone.

Previous works have shown that the TR has unique advantages in long-range context modeling. Nonetheless, TR-based methods have the drawbacks of large model size and high computational complexity. In addition, recent methods have mainly used the TR blocks to perform the same global self-attention of features in token space without considering high and low frequencies of difference features, which can easily result in high-frequency burrs and low compactness in predicted changed objects. To overcome these problems, this article aims to achieve the best balance between performance and efficiency. Therefore, an efficient backbone is designed by adopting the first three layers of the ResNet 18 to provide local prior knowledge for a lightweight TR branch. Moreover, considering the HiLo frequencies in the deep semantic difference features, the STEM is proposed for feature enhancement. Finally, three TL-FFMs are proposed instead of stand decoders to achieve an efficient feature refinement.

III. Methodology

A. Overview

In this section, the overall structure of our HSSENet is introduced in detail, and it is shown in Fig. 1. Our HSSENet utilizes a typical U-shape encoder–decoder architecture, which consists of three main components: a hybrid Siamese backbone, a STEM, and TL-FFMs. In the encoder, a pair of bitemporal images are fed to a weight-sharing hybrid backbone composed of CNN- and TR-based feature extraction branches to generate multiscale features ($\mathbf{F}^1$ and $\mathbf{F}^2$). Then, the first

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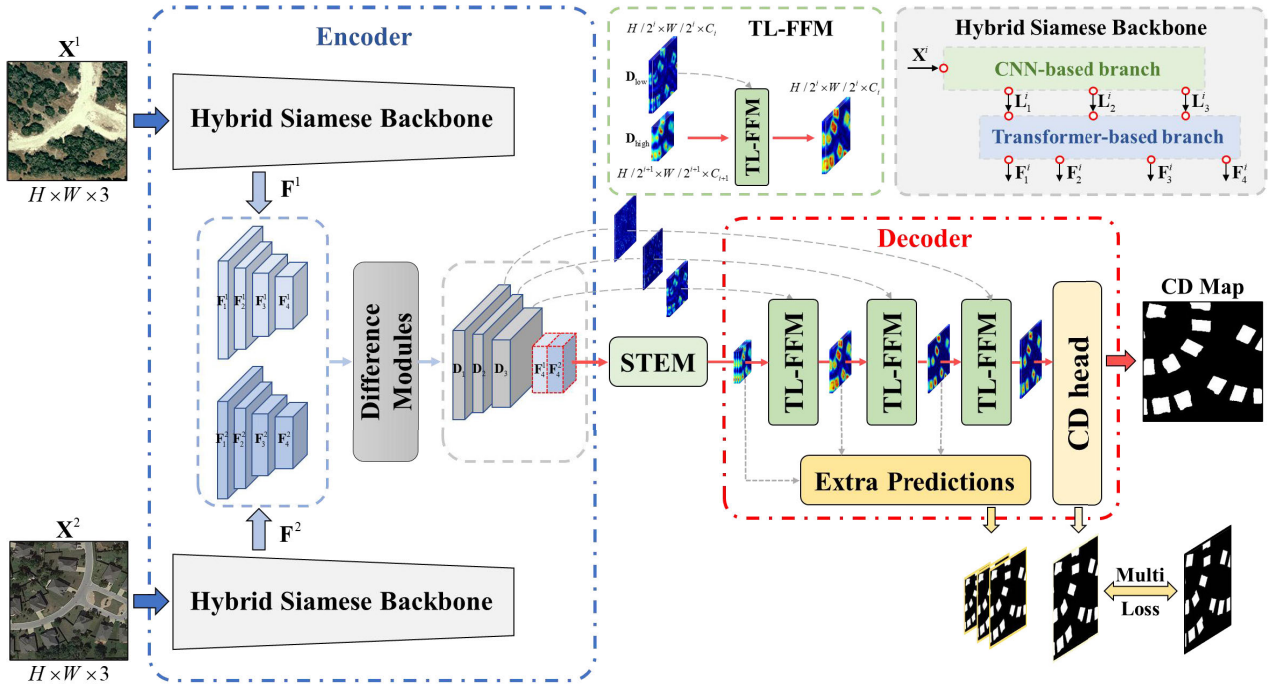


Fig. 1. Illustration of the proposed HSSENet. $\mathbf{X}^1$ and $\mathbf{X}^2$ denote the input images, which are fed in a weight-sharing hybrid backbone to generate $\mathbf{F}^1$ and $\mathbf{F}^2$, respectively; $\mathbf{D}_1$, $\mathbf{D}_2$, and $\mathbf{D}_3$ are multiscale difference features; and $\mathbf{F}_4^1$ and $\mathbf{F}_4^2$ are sent to the STEM to enhance difference feature representations; the TL-FFMs and CD head are adopted for decoding.

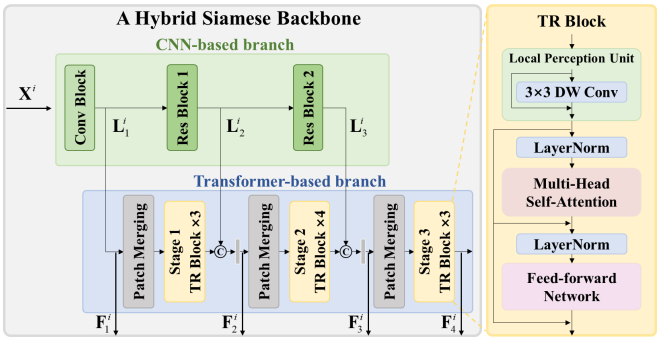


Fig. 2. Structure of the hierarchical hybrid Siamese backbone.

three of $\mathbf{F}^i$ $(i = 1, 2)$ are used to generate multiscale difference features by difference modules, which consist of channel concatenation and a 3×3 convolution operator. Meanwhile, the last level features ($\mathbf{F}_4^1$ and $\mathbf{F}_4^2$) retrieved by the feature extraction backbone are used by the STEM to enhance the difference feature representations in HiLo frequencies. Further, the three TL-FFMs are integrated into the decoder to fuse different semantic level features from decoding layers and encoding layers, and progressively refine the difference features representations. Finally, the extra predication heads and special CD head are designed to obtain multiscale predictions, and a multiloss function is employed to compute the loss between them and the corresponding ground truth.

B. Hierarchical Hybrid Siamese Backbone

Traditional CNNs have a strong inductive bias due to the translation/rotation invariance and local RF, so they can extract features using a simple sliding window with a low computational cost. However, due to the limited RF, it is difficult to obtain global contextual relationships between features. For comparison, the standard TR is suitable for context modeling, but it requires calculating the correlation between any two pixels in the feature maps, which suffers from high computational costs for HR RS images, particularly in dense prediction tasks. To combine the advantages of CNNs and TRs, this study design a hierarchical Siamese backbone. As shown in Fig. 2, our insight is that a lightweight CNN branch can provide local prior knowledge for the main TR branch, compensate for the lack of visual induction bias of TR, and effectively accelerate model training and convergence processes. In addition, using the mix of CNNs and TR can provide a better tradeoff between the model performance and efficiency. Finally, the hybrid backbone can efficiently output four feature maps: $\mathbf{F}_1^i \in \mathbb{R}^{H/2\times W/2\times 64}$, $\mathbf{F}_2^i \in \mathbb{R}^{H/4\times W/4\times 64}$, $\mathbf{F}_3^i \in \mathbb{R}^{H/8\times W/8\times 128}$, and $\mathbf{F}_4^i \in \mathbb{R}^{H/16\times W/16\times 256}$ $(i = 1, 2)$, where i represents the phase of an input image. H and W denote the height and width of input images, respectively.

The CNN-based branch is an improvement from the ResNet 18 [59]. To improve the efficiency of the backbone, this study uses only the first three layers, each with a downsampling rate of two, so we call it Res-L3 (see Section IV-E for details). The input of this branch is a preperiod image $\mathbf{X}^1 \in \mathbb{R}^{H\times W\times 3}$ or a postperiod image $\mathbf{X}^2 \in \mathbb{R}^{H\times W\times 3}$, and the outputs are multiscale semantic maps: $\mathbf{L}_1^i \in \mathbb{R}^{H/2\times W/2\times 64}$, $\mathbf{L}_2^i \in \mathbb{R}^{H/4\times W/4\times 64}$, and $\mathbf{L}_3^i \in \mathbb{R}^{H/8\times W/8\times 128}$ $(i = 1, 2)$. The TR branch is the main part of the backbone, and it consists of patch merging and TR block. The patch merging applies 3×3 convolution to achieve two-time downsampling and overlapped patch merging. Unlike the TR block in the

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vision transformer (Vit) [60] and SegFormer [61], our custom TR block contains a local perception unit (LPU), multihead self-attention module (MSAM), and feed-forward network (FFN).

Considering that Vits ignore the local feature correlation and the structure information inside the patch [62], this study not only uses the CNN branch to insert local prior knowledge into each stage of the TR branch but also designs the LPU in each TR block, which adopts a 3×3 depthwise convolution (DWConv) to introduce the translation invariance to each TR block of a stage. The residual connections are applied to enhance network stability, and the LPU$(\cdot)$ is defined as follows:

$$\mathbf{F} = \mathrm{LPU}(\mathbf{X}_\mathrm{P}) = \mathrm{DWConv}(\mathbf{X}_\mathrm{P}) + \mathbf{X}_\mathrm{P}. \tag{1}$$

In the ith TR block of each stage, the input $\mathbf{X}_\mathrm{P} \in \mathbb{R}^{H_i \times W_i \times C_i}$ is the output of patch merging or previous TR block, as shown in Fig. 2. H_i, W_i, and C_i are the height, width, and channel dimension of the current layer, respectively. DWConv$(\cdot)$ denotes the depthwise convolution, which independently performs convolution on each channel dimension of $\mathbf{X}_\mathrm{P}$ to reduce the amount of computation [63], [64].

After the LPU$(\cdot)$, $\mathbf{F} \in \mathbb{R}^{H_i \times W_i \times C_i}$ is reshaped to $\mathbf{F}' \in \mathbb{R}^{H_i \cdot W_i \times C}$ and then fed in the MSAM. In the original self-attention mechanism, the input of MSAM is linearly projected to the query $\mathbf{Q} \in \mathbb{R}^{N \times d_k}$, key $\mathbf{K} \in \mathbb{R}^{N \times d_k}$, and value $\mathbf{V} \in \mathbb{R}^{N \times d_v}$. N is the sequence length of the current stage, and it is equal to $H_i \times W_i$. d denotes the input dimension. $\mathbf{Q}$ and $\mathbf{K}$ have the same dimensions d_k. Based on the scaled dot-product mechanism, the self-attention function can be expressed as follows:

$$\mathrm{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{Soft\,max}\left(\frac{\mathbf{Q}\mathbf{K}^\mathrm{T}}{\sqrt{d_k}}\right)\mathbf{V}. \tag{2}$$

In addition, to reduce computation complexity, this study employs the convolution layer combined with a reduction ratio R to reduce the sequence lengths of $\mathbf{K}$ and $\mathbf{V}$ before self-attention. The conversion formulas are as follows:

$$\mathbf{K}' = \mathrm{Reshape}_2\big(\mathrm{Conv}\big(\mathrm{Reshape}_1(\mathbf{K})\big)\big) \in \mathbb{R}^{N/R^2 \times d_k} \tag{3}$$

$$\mathbf{V}' = \mathrm{Reshape}_2\big(\mathrm{Conv}\big(\mathrm{Reshape}_1(\mathbf{V})\big)\big) \in \mathbb{R}^{N/R^2 \times d_k} \tag{4}$$

where $\mathrm{Reshape}_1(\cdot)$ indicates the transformation from $\mathbb{R}^{N \times d_k}$ to $\mathbb{R}^{H_i \times W_i \times d_k}$, and $\mathrm{Reshape}_2(\cdot)$ indicates the opposite transformation.

The kernel size and the stride in Conv$(\cdot)$ are equal to R. After Conv$(\cdot)$, the channel dimension of feature maps remains unchanged, and only the resolution changes to $1/R$ of the original one. In the proposed HSSENet, R is set to [1, 2, 4] in TR stages 1–3. Consequently, the efficient self-attention can be defined by

$$\mathrm{EfficientAttn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{Soft\,max}\left(\frac{\mathbf{Q}\mathbf{K}'^\mathrm{T}}{\sqrt{d_k}}\right)\mathbf{V}'. \tag{5}$$

Multihead self-attention has evolved from self-attention, and the term "multihead" means that the $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ matrices are divided into h groups (heads) through multiple groups with different linear projection layers. Each head participates in a self-attention computation and outputs a sequence with a size of $n \times (d/h)$. Finally, h sequences are concatenated into an $n \times d$ sequence. Compared with single-head attention, multihead attention can jointly focus on the information of different representation subspaces from different locations [57] and improve context modeling capabilities without increasing the computational overhead. The MSA can be formulated as

$$\begin{aligned}\mathrm{MSA}(\mathbf{X}_\mathrm{L}) &= \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)\mathbf{W}^\mathrm{o} \\ \text{where } \mathrm{head}_j &= \mathrm{EfficientAttn}\big(\mathbf{X}_\mathrm{L}\mathbf{W}_j^\mathrm{q}, \mathbf{X}_\mathrm{L}\mathbf{W}_j^\mathrm{k}, \mathbf{X}_\mathrm{L}\mathbf{W}_j^\mathrm{v}\big)\end{aligned} \tag{6}$$

where the input $\mathbf{X}_\mathrm{L} \in \mathbb{R}^{N \times C}$ is the output of LPU and layer normalization (LN). h represents the number of "multihead." $\mathbf{W}_j^\mathrm{q}$, $\mathbf{W}_j^\mathrm{k}$, $\mathbf{W}_j^\mathrm{v} \in \mathbb{R}^{C \times C/h}$, and $\mathbf{W}^\mathrm{o} \in \mathbb{R}^{C \times C}$ are the linear projection matrices. Note that h is set to [2, 4, 8] in the TR stages 1–3, respectively.

Xie et al. [61] pointed out that the positional encoding (PE) in the TR is not necessary for a dense prediction task and can be replaced by inserting a simple 3×3 convolution into the FFN, which is sufficient to provide positional information for TRs. Inspired by this, this study does not use the PE in (5), instead designs two linear layers along with a 3×3 DWConv to form the FFN as follows:

$$\mathrm{FFN}(\mathbf{X}_\mathrm{M}) = \mathrm{GELU}(\mathrm{DWConv}(\mathbf{X}_\mathrm{M}\mathbf{W}_1))\mathbf{W}_2 \tag{7}$$

where the input $\mathbf{X}_\mathrm{M}$ is the output of the MSA and LN, and the dimensionality of $\mathbf{X}_\mathrm{M}$ is C. $\mathbf{W}_1 \in \mathbb{R}^{C \times 4C}$ and $\mathbf{W}_2 \in \mathbb{R}^{4C \times C}$ represent the linear projection matrices in the FFN. GELU$(\cdot)$ is the Gaussian error linear unit [65] activation.

C. Spatiotemporal Enhancement Module

At this point, high-quality hierarchical multilevel features are obtained for each temporal image. We believe that the structural integrity and the stability of changed objects will be destroyed by adopting simple subtraction operations to generate difference features. Motivated by the results presented in [57] and [60] and based on self- and cross-attention mechanisms, this study designs the STEM. The last-level outputs of the hybrid backbone, $\mathbf{F}_4^1 \in \mathbb{R}^{H/16 \times W/16 \times 256}$ and $\mathbf{F}_4^2 \in \mathbb{R}^{H/16 \times W/16 \times 256}$, are sent to the STEM for context modeling in spatiotemporal dimensions and difference feature enhancement in HiLo frequencies. As shown in Fig. 3, $\mathbf{D}_4$ is the original difference feature. $\mathbf{F}_z$ is the enhanced difference feature.

1) Spatiotemporal Interaction: The spatiotemporal interaction (STI) aims to obtain context-rich difference features $\mathbf{f}_z$ through the interaction of bitemporal features in spatiotemporal dimensions. Then, $\mathbf{f}_z$ is sent to the HiLo frequencies enhancement (HLFE) for difference feature enhancement and further reduction of irrelevant changes. Similar to the TR block in the NLP, bitemporal semantic representations $\mathbf{F}_4^1$ and $\mathbf{F}_4^2$ are separately transformed into two sets of tokens denoted by $\mathbf{Z}^1$ and $\mathbf{Z}^2$, respectively. Then, due to the powerful capability of the TR long-range modeling, contexts can be modeled in both spatial and temporal dimensions by performing self-attention. Following the standard TR block design in the vision filed [60], [61], the self-attention block in the STI is also

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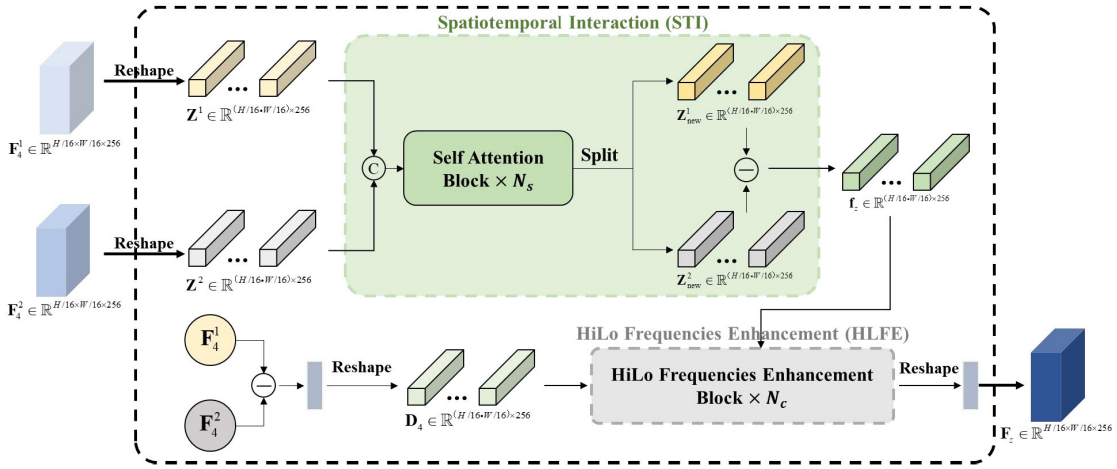


Fig. 3. Illustration of the proposed STEM. $\mathbf{F}_4^1$ and $\mathbf{F}_4^2$ are the last-level outputs of the hybrid backbone; N_s and N_c denote the numbers of self-attention and HLFE blocks, respectively; $\mathbf{D}_4$ is the original difference feature; and $\mathbf{f}_z$ denotes the context-rich difference features after STI.

composed of efficient MSA and FFN. More details of the MSA and FFN can be found in Section III-B.

As shown in Fig. 3, $\mathbf{F}_4^1$ and $\mathbf{F}_4^2$ are sent to self-attention blocks for STI

$$\mathbf{d}_z = \mathrm{MSA}(LN(\mathbf{d}_{\mathrm{in}})) + \mathbf{d}_{\mathrm{in}} \tag{8}$$

$$\mathbf{y}_z = \mathrm{FFN}(\mathrm{LN}(\mathbf{d}_z)) + \mathbf{d}_z \tag{9}$$

where the input $\mathbf{d}_{\mathrm{in}}$ represents the concatenation result of $\mathbf{F}_4^1$ and $\mathbf{F}_4^2$, and LN$(\cdot)$ denotes the LayerNorm function.

After N_c self-attention blocks, $\mathbf{Y}_z \in \mathbb{R}^{(H/16\cdot W/16)\times 512}$ can be obtained. Next, $\mathbf{Y}_z$ is split into two new sets of tokens denoted by $\mathbf{Z}_{\mathrm{new}}^1 \in \mathbb{R}^{(H/16\cdot W/16)\times 256}$ and $\mathbf{Z}_{\mathrm{new}}^2 \in \mathbb{R}^{(H/16\cdot W/16)\times 256}$, and $\mathbf{f}_z$ is the elementwise absolute of their subtraction, which is expressed as follows:

$$\mathbf{Z}_{\mathrm{new}}^1, \mathbf{Z}_{\mathrm{new}}^2 = \mathrm{split}(\mathbf{Y}_z) \tag{10}$$

$$\mathbf{f}_z = |\mathbf{Z}_{\mathrm{new}}^1 - \mathbf{Z}_{\mathrm{new}}^2|. \tag{11}$$

2) HiLo Frequencies Enhancement: Similar to the self-attention block in the STI [as in (8) and (9)], the cross-attention block in the HLFE replaces the MSA with the efficient multihead cross-attention (MCA). The HLFE is developed considering that low frequencies in image features mostly capture global structure information about targets, while high frequencies contain high-frequency noise and fine-grained details of the targets [66]. In the RS image CD field, only the low-frequency part of the overall structure and high-frequency boundary details of targets are worth attention, while the high-frequency noise and irrelevant change interference caused by the subtraction of bitemporal features should be suppressed. Therefore, the HLFE deals with low and high frequencies separately, thus achieving an effective difference feature enhancement in different frequencies. As shown in Fig. 4, in low-frequencies enhancement (LoFE) part, since averaging is low-pass filtering [67], first, average pooling is applied to $\mathbf{D}_4$ to obtain its low-frequency signals ($\mathbf{D}'_4 \in \mathbb{R}^{((H/32)\cdot(W/32))\times 256}$) which can help to enhance the internal compactness of changed objects. In high-frequencies enhancement (HiFE) part, $\mathbf{f}_z \in \mathbb{R}^{((H/16)\cdot(W/16))\times 256}$ denotes the

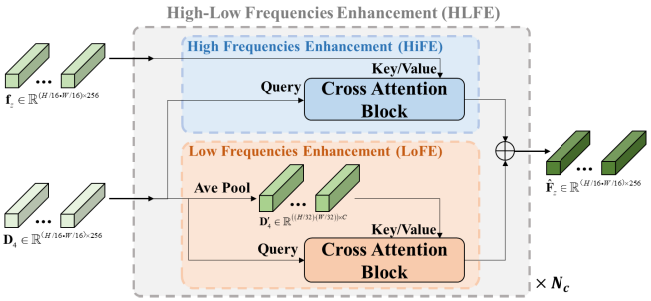


Fig. 4. Illustration of our HLFE. N_c denotes the number of high–low frequencies enhancement block.

context-rich difference feature with greater spatial resolution derived by the STI. Therefore, it is thought to contain more relatively valuable high-frequency information about changed objects. It can help to suppress irrelevant pseudo-changes and noise interference in $\mathbf{D}_4$ and maintain the boundary profile of a target.

Therefore, the HiFE uses $\mathbf{D}_4$ as the query and $\mathbf{f}_z$ as the key and value. Accordingly, the global relationships between each query position in $\mathbf{D}_4$ and the context-rich high-frequency key in $\mathbf{f}_z$ can be modeled. This can reduce high-frequency noise interference and eliminate irrelevant small changes in target objects and only focus on sharp edges of the targets. Another branch (LoFE) also uses $\mathbf{D}_4$ as the query, but it employs $\mathbf{D}'_4$ as the key and value to perform the MCA. This can help to enhance the rich low-frequency information in difference features, such as the overall structural representations of changed targets. After N_c times of the HiFE and LoFE, the outputs of two branches are fused, and a 1×1 Conv-batch normalization-rectified linear unit (BN-ReLU) layer is used to obtain an enhanced difference feature $\mathbf{F}_z$. The MCA expressions of the HiFE and LoFE with one block are defined by the following equations, respectively:

$$\begin{aligned} &\mathrm{MCA}_h(\mathbf{D}_4, \mathbf{f}_z) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)\mathbf{W}^{\mathrm{o}} \\ &\text{where head}_j = \mathrm{EfficientAttn}\left(\mathbf{D}_4\mathbf{W}_j^{\mathrm{q}}, \mathbf{f}_z\mathbf{W}_j^{\mathrm{k}}, \mathbf{f}_z\mathbf{W}_j^{\mathrm{v}}\right) \end{aligned} \tag{12}$$

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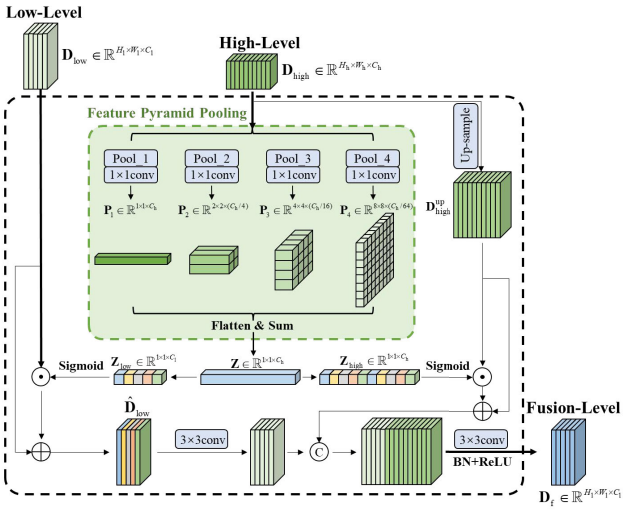


Fig. 5. Details of our TL-FFM.

$$\begin{aligned}&\mathrm{MCA}_l(\mathbf{D}_4, g(\mathbf{D}_4))\\&\quad=\mathrm{Concat}(\mathrm{head}_1,\ldots,\mathrm{head}_h)\mathbf{W}^{\mathrm{o}}\\&\text{where head}_j=\mathrm{EfficientAttn}\left(\mathbf{D}_4\mathbf{W}_j^{\mathrm{q}}, g(\mathbf{D}_4)\mathbf{W}_j^{\mathrm{k}}, g(\mathbf{D}_4)\mathbf{W}_j^{\mathrm{v}}\right).\end{aligned}\tag{13}$$

In (12) and (13), $\mathbf{D}_4=\mathrm{Reshape}_2(\Phi(|\mathbf{F}_4^1-\mathbf{F}_4^2|))$ is generated by implementing the elementwise absolute of the subtraction on $\mathbf{F}_4^1$ and $\mathbf{F}_4^2$. $\Phi(\cdot)$ represents a 1×1 Conv-BN-ReLU layer. $g(\cdot)$ has a similar effect to the average pooling. Consequently, the output ($\mathbf{F}_z$) of the STEM is obtained by (N_c is set to 1)

$$\hat{\mathbf{F}}_z=\mathrm{HLFE}(\mathbf{D}_4,\mathbf{f}_z)=\mathrm{high}+\mathrm{low}\tag{14}$$

$$\mathbf{F}_z=\Psi\left(\mathrm{Reshape}_1\left(\hat{\mathbf{F}}_z\right)\right)\tag{15}$$

where high and low denote the HiFE and LoFE branches' outputs, respectively; $\Psi(\cdot)$ is a 1×1 Conv-BN-ReLU layer, which is used to improve model robustness. It should be noted when $N_c>1$, $\hat{\mathbf{F}}_z^{(i)}$ is the output of the ith HLFE block, and it is also used as $\mathbf{D}_4^{(i+1)}$ in the $(i+1)$th HLFE block.

D. Two-Level Feature Fusion Module

Inspired by the pyramid pooling [68] and unequal channel pyramid pooling [69] in the semantic segmentation task, we believe that information on multiscale target features is vital for accurate CD tasks. Also, it is considered that the refinement of multiscale targets in the decoder is important to the feature extraction in the encoder. Therefore, based on the improved pyramid pool module and channel attention mechanism, efficient two-stage feature fusion modules are designed for decoding in the RS image CD. The conventional decoder often directly aggregates the upstream and downstream feature maps, thus losing spatial details, which can lead to ambiguity for changed objects. Different from that, the proposed TL-FFM uses high-level semantic-rich feature representations to guide the channel weighting of low- and high-level feature representations, as illustrated in Fig. 5. Consequently, the loss in local information in the deep layers of the proposed model can be reduced, and the boundary information can be refined. In addition, three TL-FFMs are used for decoding, and in each of them, the high-level feature $\mathbf{D}_{\mathrm{high}}\in\mathbb{R}^{H_{\mathrm{h}}\times W_{\mathrm{h}}\times C_{\mathrm{h}}}$ is obtained by the STEM or the previous TL-FFM, and the low-level feature $\mathbf{D}_{\mathrm{low}}\in\mathbb{R}^{H_{\mathrm{l}}\times W_{\mathrm{l}}\times C_{\mathrm{l}}}$ is obtained by the difference module ($H_{\mathrm{l}}=2\times H_{\mathrm{h}}$, $W_{\mathrm{l}}=2\times W_{\mathrm{h}}$, and $C_{\mathrm{l}}=C_h/2$). For instance, in the first TL-FFM (see Fig. 1), $\mathbf{D}_{\mathrm{high}}$ corresponds to $\mathbf{D}_4$, and $\mathbf{D}_{\mathrm{low}}$ corresponds to $\mathbf{D}_3$.

Particularly, $\mathbf{D}_{\mathrm{high}}$ is fed to the improved pyramid pooling and subject to four different scale average pooling operations and 1×1 convolution to obtain difference features with multiscale contextual information parallelly: $\mathbf{P}_1\in\mathbb{R}^{1\times1\times C_{\mathrm{h}}}$, $\mathbf{P}_2\in\mathbb{R}^{2\times2\times(C_{\mathrm{h}}/4)}$, $\mathbf{P}_3\in\mathbb{R}^{4\times4\times(C_{\mathrm{h}}/16)}$, and $\mathbf{P}_4\in\mathbb{R}^{8\times8\times(C_{\mathrm{h}}/64)}$; $\mathbf{P}_1$ contains global semantic information, and the other three contain local details of subregions with different scales. Different from regular pyramid pooling, in the proposed improved pyramid pooling module, features are compressed in both spatial and channel dimensions. By flattening and summing $\mathbf{P}_1$, $\mathbf{P}_2$, $\mathbf{P}_3$, and $\mathbf{P}_4$, the weight vector $\mathbf{Z}\in\mathbb{R}^{1\times1\times C_{\mathrm{h}}}$ can be obtained. $\mathbf{Z}$ is linearly transformed to $\mathbf{Z}_{\mathrm{low}}\in\mathbb{R}^{1\times1\times C_{\mathrm{l}}}$ and $\mathbf{Z}_{\mathrm{high}}\in\mathbb{R}^{1\times1\times C_{\mathrm{h}}}$, which can be used for channel weighting selection of $\mathbf{D}_{\mathrm{low}}$ and $\mathbf{D}_{\mathrm{high}}^{\mathrm{up}}$, respectively. The corresponding mathematical expressions are as follows:

$$\mathbf{Z}_{\mathrm{low}}=\delta(\mathrm{Linear}_{\mathrm{low}}(\mathbf{Z}))\tag{16}$$

$$\mathbf{Z}_{\mathrm{high}}=\delta\left(\mathrm{Linear}_{\mathrm{high}}(\mathbf{Z})\right)\tag{17}$$

$$\hat{\mathbf{D}}_{\mathrm{low}}=\mathbf{Z}_{\mathrm{low}}\odot\mathbf{D}_{\mathrm{low}}+\mathbf{D}_{\mathrm{low}}\tag{18}$$

$$\hat{\mathbf{D}}_{\mathrm{high}}^{\mathrm{up}}=\mathbf{Z}_{\mathrm{high}}\odot\mathbf{D}_{\mathrm{high}}^{\mathrm{up}}+\mathbf{D}_{\mathrm{high}}^{\mathrm{up}}\tag{19}$$

where $\mathbf{D}_{\mathrm{high}}^{\mathrm{up}}\in\mathbb{R}^{H_{\mathrm{l}}\times W_{\mathrm{l}}\times C_{\mathrm{h}}}$ denotes the upsampled high-level features, and it has the same spatial resolution as $\mathbf{D}_{\mathrm{low}}$. $\mathbf{Z}_{\mathrm{low}}$ and $\mathbf{Z}_{\mathrm{high}}$ are the weight vectors for channel weighting selection. $\mathrm{Linear}_{\mathrm{low}}(\cdot)$ and $\mathrm{Linear}_{\mathrm{high}}(\cdot)$ are the linear projections. $\delta(\cdot)$ is a sigmoid function for limiting the weight vectors to the interval of [0, 1]. $\odot$ represents the elementwise multiplication.

A 3×3 Conv is employed on the enhanced low-level difference feature representations to improve model robustness, which is followed by the fusion of HiLo-level features, thus yielding to:

$$\mathbf{D}_{\mathrm{f}}=\Theta\left(\mathrm{Concat}\left(\mathrm{Conv}\left(\hat{\mathbf{D}}_{\mathrm{low}}\right),\hat{\mathbf{D}}_{\mathrm{high}}^{\mathrm{up}}\right)\right)\tag{20}$$

where $\Theta(\cdot)$ denotes a 3×3 Conv-BN-ReLU layer. $\mathbf{D}_{\mathrm{f}}$ is the TL-FFM output, and it has the same dimension as $\mathbf{D}_{\mathrm{low}}$. It should be noted that the output $\mathbf{D}_{\mathrm{f}}$ of the previous TL-FFM will be used as $\mathbf{D}_{\mathrm{high}}$ of the next TL-FFM.

E. Multilevel Extra Predictions and CD Head

The multilevel extra predictions are proposed, so that the most representative difference features of different semantic levels can all be involved in guiding model training. As shown in Fig. 6, the extra prediction module takes the outputs of the STEM and first two TL-FFMs as inputs. The 3×3 Conv-BN-ReLU-3×3 Conv layer is applied to each input to generate the multiscale CD maps. In addition, we believe that a simple upsampling operation has limited capability in recovering pixel prediction and may easily result in losing the edge details of the changed targets. Furthermore, TL-FFMs lightly bring some high-frequency interferences due to the aggregation of HR raw information. As shown in the right side in Fig. 6, a special

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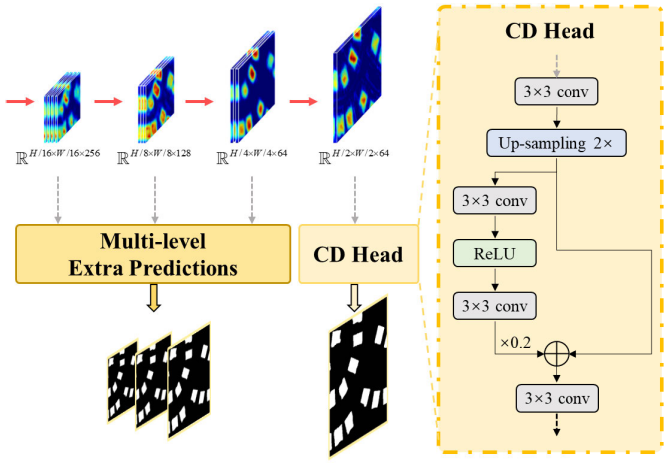


Fig. 6. Details of the extra predictions and CD head. The top four groups of the feature maps represent the outputs of the STEM and three TL-FFMs.

CD head is designed, which acts on the last-level difference features. The 3 × 3 Conv and skip connection are added after upsampling to filter interference and stabilize the postsampling performance.

F. Multiloss Function

Similar to the semantic segmentation task, the RS image CD can essentially be regarded as a pixel-level classification task; hence, we minimize the cross-entropy (CE) loss to optimize model parameters. In view of additional multilevel predictions, the multiloss function can be defined by

$$L = \sum_{i=1}^{K} \lambda_i \left(\frac{1}{H_i \times W_i} \sum_{h=1,w=1}^{H_i,W_i} l(P_{\text{hw}}, Y_{\text{hw}}) \right) \tag{21}$$

where K denotes the number of total predictions; λ_i is the weight of the ith level prediction, and the smaller its subscript is, the farther it is to the final output of the CD model; H_i and W_i are the height and width of the ith-level prediction; $l(P_{hw}, Y_{hw}) = -\log(P_{hwy})$ is the CE loss; P_{hw} is the prediction after a softmax function at a location (h, w); and Y_{hw} is the corresponding label. In experiments, K was set to 4, and $\lambda = [\lambda_1, \lambda_2, \lambda_3, \lambda_4]$ was set to $[1, 1, 1, 1]$ for four multilevel predictions. The labels were resized to the same size as the predicted CD maps in the nearest manner, and then, the loss between them was calculated. Four multilevel predictions were used for training, but only the last-level prediction was used as a final inference result on the test sets.

IV. Experiments and Results

A. Datasets Description

To demonstrate the effectiveness of the proposed method, extensive experiments were conducted on the learning, vision, and RS (LEVIR) [44], Wuhan University (WHU) [8], and deeply supervised image fusion network (DSIFN) [70] datasets.

1) The LEVIR dataset is a public large-scale building CD dataset collected from the Google Earth (GE) platform, including 637 pairs of HR (0.5 m) optical satellite RS images of a size of 1024 × 1024. Due to the GPU memory capacity limitation, the data were split, and a nonoverlapping sliding window was used to cut each image into 16 256 × 256 image slices. Then, the obtained data were divided into training, validation, and test sets containing 7120, 1024, and 2048 pairs of image patches, respectively.
2) The WHU dataset is a public building CD dataset, which contains a pair of HR (0.075 m) aerial RS images with a size of 32 507 × 15 354. Since no data division method was provided in [8], the general data division method [46] was performed on this dataset. A nonoverlapping sliding window was used to cut the HR image into slices with a size of 256 × 256, and they were randomly split into training, validation, and test sets containing 6096, 762, and 762 pairs of images, respectively.
3) The DSIFN dataset is a public CD dataset that contains six large pairs of HR images (2 m) collected by the GE platform in the area covering six different cities in China. Since these images were collected by various sensors, there are great differences in image contrast and brightness between the images in this dataset, which poses challenges to the CD task. Unlike the first two datasets, the DSIFN dataset contains various changes in land cover, such as buildings, roads, and farmland. Consistent with the first two division principles, the images were cropped into 256 × 256 pieces in a nonoverlapping way, and the obtained data were split into training, validation, and test sets containing 14 400, 1360, and 192 pairs of images, respectively.

B. Experimental Setup

The model performance was evaluated by experiments on the three datasets (LEVIR, WHU, and DSIFN datasets). In addition to eight state-of-the-art (SOTA) models as a comparison, the proposed model settings were as follows.

HSSENet: Our hybrid backbone consisted of the Res-L3 and TR-L3 (see Section IV-E for details) + STEM + TL-FFMs. The Res-L3 means that the CNN branch in the backbone consists of the first three stages of the ResNet18. The TR-L3 denotes that the TR branch in the backbone consists of three stages, each with a downsampling rate of two.

1) Evaluation Metrics: This study adopted Pr, Re, OA, F_1, and IoU metrics to evaluate the performance of the proposed algorithm. Metrics F_1 and IoU represent the F1-score and intersection over union, respectively. These two metrics are comprehensive evaluation indicators, and they denote the main performance indicators in this study. The larger their values are, the better the model performance is. Metrics Pr, Re, and OA represent the precision rate, recall rate, and overall accuracy, respectively. The evaluation metrics were calculated as follows:

$$\text{Pr} = \text{TP}/(\text{TP} + \text{FP})$$
$$\text{Re} = \text{TP}/(\text{TP} + \text{FN})$$
$$\text{OA} = (\text{TP} + \text{TN})/(\text{TP} + \text{FP} + \text{TN} + \text{FN})$$

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$$F_1 = 2\mathrm{Pr} \cdot \mathrm{Re}/(\mathrm{Pr} + \mathrm{Re})$$
$$\mathrm{IoU} = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP} + \mathrm{FN}) \quad (22)$$

where TP, FP, FN, and TN are the numbers of true positives, false positives, false negatives, and true negatives, respectively.

2) Implementation Details: Our models were constructed under the PyTorch DL framework and trained on a single NVIDIA RTX 3090 GPU (24-GB RAM). The normal data augmentation method was used during the training stage, including random flip, rescale, crop, Gaussian blur, and random color jittering. The CE loss and AdamW optimizer were used to train and optimize the models. The weight decay was set to 0.01, and the beta values were in the range of (0.9, 0.999). The learning rate was initially set to 0.0001, and it linearly decayed to zero after 200 training epochs. The batch size was set to eight. Noted that the SOTA methods were constructed according to their original parameters.

C. Comparison With Other Methods

1) Comparison Methods: The proposed HSSENet was compared with three commonly used CD models based on the U-Net series and five recent SOTA models in the CD task. The U-Net-based models included the FC-EF [37], FC-Siam-Diff [37], and FC-Siam-Conc [37], and they were all full-convolution CD networks based on the skip connection and feature fusion. The SOTA models included the dual-task constrained deep Siamese convolutional network (DTCDSCN) [42], TR-based Siamese network (ChangeFormer) [45], BIT-CD [46], deeply supervised image fusion network (IFNet) [70], and densely connected Siamese network (SNUNet) [41]. The DTCDSCN, IFNet, and SNUNet were based on the CNN architecture with various attention modules. The DTCDSCN adopted a dual attention module (DAM) to exploit interdependencies in the channel and spatial dimensions. Moreover, the semantic segmentation and CD tasks were integrated into the DTCDSCN, but the semantic segmentation decoders were omitted in the experiments for the purpose of a fair comparison. The IFNet applied the attention modules to improve the feature representations, and multilevel features were fused to improve the boundary completeness and structural integrity of changed targets. The SNUNet established a dense connection between the encoder and decoder, as well as between the decoder and decoder. An ensemble channel attention module was also embedded in the SNUNet. The ChangeFormer was based on pure TR architecture, and the hierarchically pure TR-based encoder was proposed for the CD task. The BIT represented a hybrid model based on CNNs and TR architecture, where a TR-based decoder was applied to model long-distance dependencies between features in the token-based space. It is worth mentioning that all methods were retrained for 200 epochs according to the relevant hyperparameters introduced in the original articles to ensure fairness.

2) Comparison on Three Datasets: The comparison experiments were conducted on the three datasets, and several challenging and representative situations of changes were selected to illustrate the results. The comparison results on the LEVIR, WHU, and DSIFN datasets are presented in Figs. 7–9. Based on the visual comparison results, the proposed HSSENet achieved better performance than the SOTA models in different situations. For instance, in situations with complex edge contours of the changed targets [e.g., Figs. 7(II)–(IV), 8(II) and (III), and 9(II) and (IV)], the proposed HSSENet could not only maintain a compact structure of objects but also compensate for the detailed information lost in the downsampling process. This could be because the proposed TL-FFMs integrated the features of different semantic levels. Further, when changes were of a small scale [e.g., Figs. 7(II) and (IV), 8(IV), and 9(II) and (IV)], the SOTA models showed different degrees of misdetection, especially in Fig. 9(II), where there can be observed a number of small changes. However, the proposed HSSENet could accurately identify all small changes and achieved advantages in terms of boundary accuracy compared to the SOTA models. This could be attributed to the powerful context modeling of the proposed hybrid Siamese backbone. In addition, when dense changes appeared [see Fig. 9(IV)], there were certain adhesions in some of the models, including the DTCDSCN and IFNet models. In contrast, the proposed HSSENet could alleviate this phenomenon to a certain extent. It should be noted that objects with the same semantic concept in complex scenes may show different spectral characteristics at different positions or different times, leading to false detection in most methods, such a situation is presented by the red circle in Fig. 7(IV). However, the proposed HSSENet could effectively avoid the false detection phenomenon because it used the STEM to model the dense relationships between semantic features of the bitemporal images in the spatiotemporal dimensions. More importantly, when the changed objects had a regular shape, the proposed model could also maintain better boundary smoothness [Figs. 8(II)–(IV) and 9(I) and (III)] and internal structural compactness [Figs. 7(I) and (III) and 8(I)] than the other models, and effectively eliminated high-frequency burrs [Figs. 8(III) and 9(I)]. This could be attributed to the HLFE in the STEM.

Therefore, the proposed model could improve the detection rate of small changes, refine the edge details of complex objects, and achieve the best performance in boundary integrity and internal compactness among all models. The quantitative analysis results are displayed in Table I to demonstrate the superiority of the proposed model. In Table I, the best values are marked in bold red, and the second values are marked in bold blue. As shown in Table I, on the three public test sets, the proposed HSSENet obtained the F1-score and IoU values of 91.48, 91.55, and 91.17 and 84.29, 84.42, and 83.77 on the LEVIR, WHU, and DSIFN datasets, respectively, achieving the best performance among all models on each dataset. The F1-score and IoU results of the proposed HSSENet exceeded those of the second-best model by 0.54, 1.58, and 3.04 and 0.91, 2.66, and 5.0 points on the LEVIR, WHU, and DSIFN datasets, respectively. In Table I, it can be noted that most comparison models had different performances on the three datasets, while our model obtained high accuracy and good robustness on all datasets, especially on the complex DSIFN dataset.

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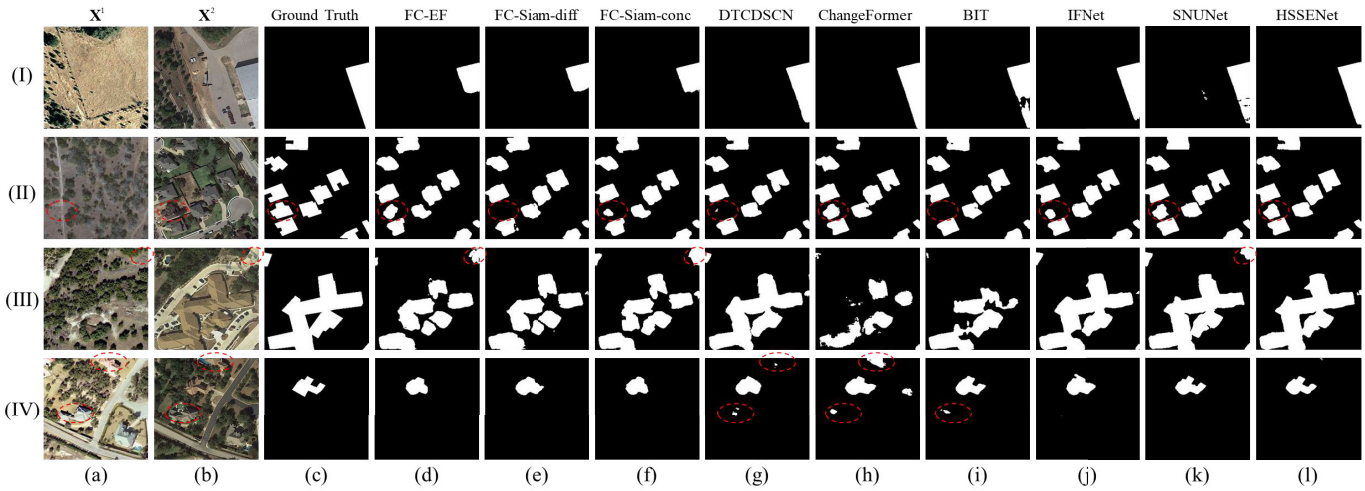


Fig. 7. Visual comparison results of various situations on the LEVIR test sets. Pseudo-changes and misdetections in complex scenes are shown in red circles. (a)–(c) Preperiod image, postperiod image, and label. (d)–(l) Predictions of different models.

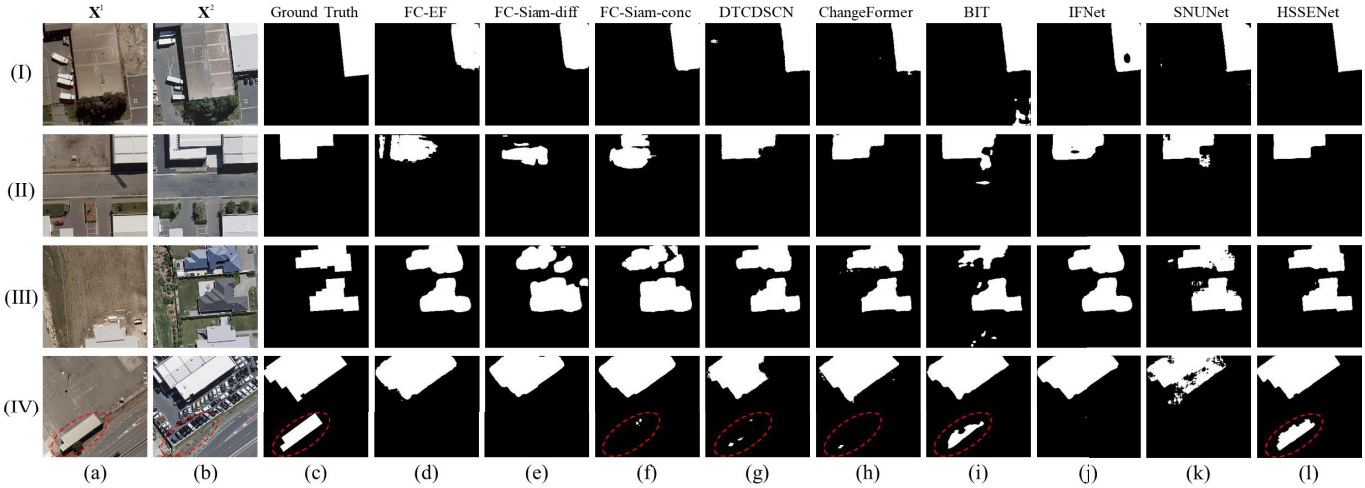


Fig. 8. Visual comparison results of various situations on the WHU test sets. The misdetections are shown in red circles. (a)–(c) Preperiod image, postperiod image, and label. (d)–(l) Predictions of different models.

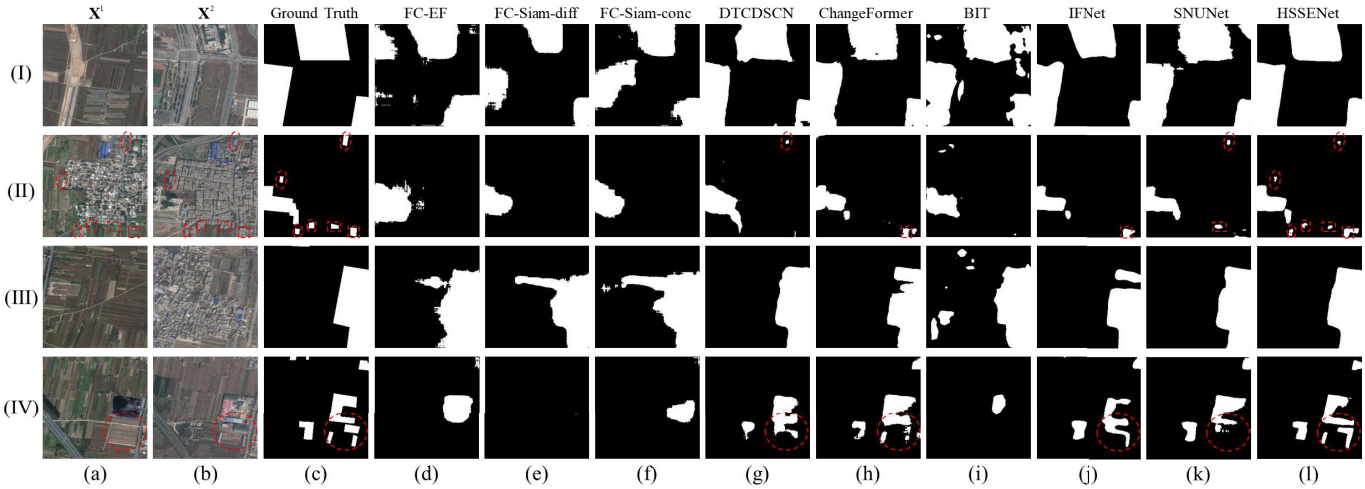


Fig. 9. Visual comparison results of various situations on the DSIFN test sets. Small changes are shown in red circles. (a)–(c) Preperiod image, postperiod image, and label. (d)–(l) Predictions of different models.

3) Efficiency Comparison: The proposed HSSENet aims to leverage the advantages of both the CNN and the TR to obtain a tradeoff between the performance and overhead, so the models were analyzed from multiple perspectives, such as the number of model parameters (Params), the number of floating-point operations (FLOPs), and the inference time.

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TABLE I

COMPARISON RESULTS OF THE LEVIR, WHU, AND DSIFN TEST SETS. THE BEST VALUES ARE MARKED IN BOLD RED, AND THE SECOND-BEST VALUES ARE MARKED IN BOLD BLUE; ALL SCORES ARE EXPRESSED IN %

Method	LEVIR-CD	WHU-CD	DSIFN-CD
	Pr / Re / F1 / IoU / OA	Pr / Re / F1 / IoU / OA	Pr / Re / F1 / IoU / OA
FC-EF[37]	86.09 / 83.86 / 84.96 / 73.85 / 98.49	84.66 / 70.85 / 77.14 / 62.79 / 98.33	51.12 / 69.44 / 58.89 / 41.73 / 83.52
FC-Siam-Diff[37]	89.36 / 84.87 / 87.06 / 77.08 / 98.71	86.80 / 77.18 / 81.75 / 69.07 / 98.69	69.86 / 57.39 / 63.02 / 46.00 / 88.55
FC-Siam-Conc[37]	90.26 / 84.86 / 87.48 / 77.74 / 98.71	86.20 / 84.12 / 85.15 / 74.14 / 98.84	62.75 / 68.83 / 65.65 / 48.86 / 87.76
DTCDSCN[42]	91.58 / 88.82 / 90.18 / 82.12 / 99.02	93.35 / 86.82 / 89.97 / 81.76 / 99.23	86.99 / 89.15 / 87.15 / 77.10 / 95.60
ChangeFormer[45]	90.80 / 88.56 / 89.67 / 81.27 / 98.96	91.74 / 84.66 / 88.06 / 78.66 / 99.09	89.90 / 86.01 / 87.91 / 78.43 / 95.98
BIT-CD[46]	92.34 / 87.71 / 89.96 / 81.76 / 99.00	88.99 / 88.99 / 86.81 / 76.69 / 98.93	81.44 / 74.38 / 77.75 / 63.59 / 92.77
IFNet[70]	91.78 / 89.75 / 90.61 / 82.84 / 99.06	90.62 / 88.75 / 89.67 / 81.27 / 99.19	91.27 / 80.99 / 80.99 / 75.17 / 95.45
SNUNet[41]	91.92 / 89.98 / 90.94 / 83.38 / 99.09	90.04 / 86.42 / 88.20 / 78.89 / 99.08	89.21 / 87.06 / 88.13 / 78.77 / 96.01
HSSENet	92.63 / 90.35 / 91.48 / 84.29 / 99.14	94.56 / 88.73 / 91.55 / 84.42 / 99.35	92.52 / 89.86 / 91.17 / 83.77 / 97.04

TABLE II

PARAMS, FLOPS, AND INFERENCE TIME OF DIFFERENT METHODS. THE INPUT IMAGE SIZE WAS 256 × 256 × 3

Method	Params (M)	FLOPs (G) w/bs=1	Inference Time (ms) w/bs=8
FC-EF[37]	1.35	3.57	7.10
FC-Siam-Diff[37]	1.35	4.72	10.5
FC-Siam-Conc[37]	1.55	5.32	10.69
DTCDSCN[42]	41.07	13.22	19.49
ChangeFormer[45]	41.03	202.87	122.22
BIT-CD[46]	3.55	10.59	30.01
IFNet[70]	50.71	82.35	66.46
SNUNet[41]	12.03	54.88	67.28
HSSENet	**10.95**	**17.14**	**44.30**

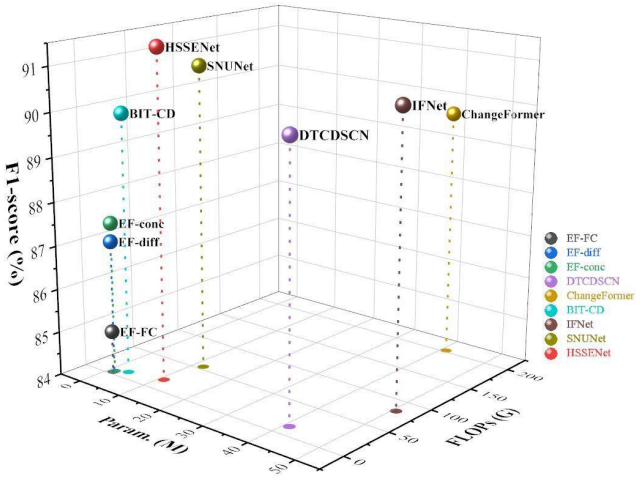


Fig. 10. Comparison results of the overhead (Param/FLOPs) versus F1-score on the LEVIR test sets.

Tests based on Table II were performed on a single NVIDIA RTX 3090 GPU (24-GB RAM). As shown in Table II and Fig. 10, the three models of the U-shape basic series (i.e., the FC-EF, FC-Siam-Diff, and FC-Siam-Conc models) had the lowest training overhead and inference time due to the shallow depth of feature extraction, which led to poor detection performance. The BIT-CD model also had relatively low detection accuracy due to its simple feature extraction and modeling semantic relevance only in the token-based space. The other comparison models achieved visible performance improvements due to the introduction of the embedded attention modules or TR blocks, having strong context modeling capabilities. However, the proposed HSSENet outperformed all models in the F1-score and IoU metrics, having only 10.95 M parameters and 17.14 G of the calculation amount, which was projected to the lower left corner of the Params–FLOPs plane in Fig. 10. The inference time of HSSENet is 44.3 ms. It was 77.92 ms faster than TR-based ChangeFormer, and 22.16 and 22.98 ms faster than IFNet and SNUNet, which are based on attention mechanisms and dense connectivity. In fact, the special design of the hybrid backbone with CNN- and TR-based branches and the efficient fusion between encoder and decoder could be the main factors for such significant improvements compared to other models.

D. Ablation Studies

The proposed HSSENet integrates a hybrid backbone consisting of the CNN- and TR-based branches, STEM, and TL-FFMs for efficient CD. Ablation experiments were conducted on the three datasets to study the contribution of each module in the proposed method to its performance. To illustrate the advantages of using the CNN and TR branches in parallel, they were tested in series in the experiments as a comparison (see Proposed 1). Since the convolution and TR computations are correlated with the channel dimension and resolution of the input features, respectively [57], only the CNN-TR in series was considered in the ablation experiments.

To sum up, the model combination configurations used in the ablation experiments were as follows. Note that CNN- and TR-branches in our backbone are used in parallel by default.

1) *Base-L5*: A modified ResNet18 + CD head. The original ResNet-18 structure was modified by setting the stride of the last two stages to one, which made the final

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output stride size is equal to 8. In addition, to implement the CD task, we introduce elementwise absolute of the subtraction and upsampling layers as the CD head. Since the ResNet-18 had a total of five stages, we named this CD model Base-L5.

2) *Proposed 1*: Res-L3 + our TR-L3 (in series). We added the upsampling layers at the end of the Res-L3 to ensure the output stride size of the CNN branch was 2, so that the two can be employed in series.
3) *Proposed 2*: Res-L3 + our TR-L3.
4) *Proposed 3*: Res-L3 + our TR-L3 + our STEM.
5) *Proposed 4*: Res-L3 + our TR-L3 + our TL-FFMs.
6) *HSSENet*: Res-L3 + our TR-L3 + our STEM + our TL-FFMs.

It can be found in Table III that replacing a modified ResNet-18 (Base-L5) with our hybrid feature extraction backbone (Proposed 1 and Proposed 2) not only reduced the number of model parameters by almost a third, and the computation by two-fifths, but it also increased the F1-score on the three test sets, especially on the DSIFN test sets. Moreover, using the CNN and TR branches in parallel provided a greater performance improvement than using them sequentially, which could be because the serial manner did not introduce a strong visual induction bias of convolution into each stage of the TR, thus leading to the limited performance of the model. This also increased the network depth of Proposed 1, which could easily cause model instability during training. The STEM (Proposed 3) further improved the performance by 0.13, 0.41, and 2.99 points of the F1-score on the LEVIR, WHU, and DSIFN datasets, respectively; the improvement was the most obvious on the complex DSIFN dataset, which could be because the context modeling in spatiotemporal dimensions and difference enhancement in HiLo frequencies helped to eliminate pseudo-changes and maintain compact structure of targets in the complex scenes. In addition, compared with Proposed 2, in Proposed 4, the three TL-FFMs were added to the decoder to fuse the internal structure information of high level and the edge detail information of low level, which further increased the F1-score value by 0.33, 0.31, and 1.43 points with a little increase in the number of parameters on the LEVIR, WHU, and DSIFN datasets. The combination of the proposed hybrid backbone, STEM, and TL-FFMs (the proposed HSSENet) significantly outperformed the Base-L5 by 1.07, 0.56, and 24.12 points in the F1-score on the LEVIR, WHU, and DSIFN datasets with a less overhead. The ablation experiments' results on the three datasets are shown in Fig. 11. The above results demonstrate that each module in the proposed HSSENet improves the performance to different degrees while maintaining a relatively small number of model parameters and a low computational cost.

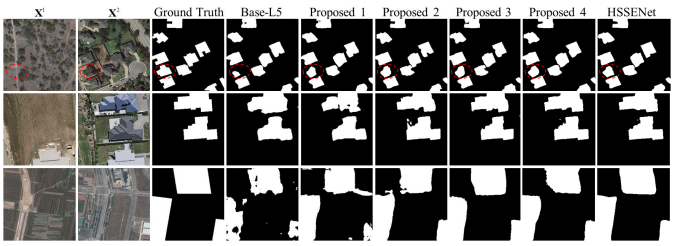


Fig. 11. Visualization results of the ablation studies on the three test sets. Lines 1–3 indicate LEVIR/WHU/DSIFN datasets, respectively.

E. Discussion

1) Discussion on the Hybrid Backbone Configuration: To explore the influence of the depth and width of the proposed hybrid Siamese backbone on the performance, we used the number of stages of the CNN-based branch (C.N) and TR-based branch (T.N) as the main hyperparameters to perform a series of experiments. The remaining hyperparameters were related to the C.N and T.N. The B.N and B.C denoted the number of output levels and the number of output channels in each stage of the backbone, respectively. The C.C and T.C were the numbers of output channels in each stage of the CNN and TR branches, respectively. The Res-L3 (C.N: 3) indicated the CNN-based branch consisting of three stages, and the meaning of Res-L4, Res-L5, TR-L3, and TR-L4 was similar. Further, TR-L3W (T.C: [128, 256, 512]) indicated that it is wider than TR-L3 (T.C: [64, 128, 256]). It should be noted that each stage of the CNN and TR branches was accompanied with a downsampling rate of two, and all variants of the hybrid backbone fed the output of their last four stages to the subsequent decoder. The number of self-attention blocks (SA.N) and the number of cross-attention blocks (CA.N) represented the depth of the STI and HLFE in STEM, respectively. They were both set to 1.

The values of C.C and T.C were increased to increase the model width, and the values of C.N and T.N were increased to enlarge the model depth. As shown in Table IV, when the backbone width was increased (i.e., "Res-L3 + TR-L3W" and "Res-L4 + TR-L3W") based on "Res-L3 + TR-L3" and "Res-L4 + TR-L3," although a slight performance improvement was achieved (0.12, 0.11, and 1.44 points and 0.11, 0.06, and 2.02 points of F1 on the LEVIR, WHU, and DSIFN datasets, respectively), it also significantly increased the number of model parameters and computational cost, which was not in line with the original aim of the proposed model. More importantly, when increasing the C.N to achieve a deeper network, there was an unexpected drop in the performance, particularly on the DSIFN dataset (i.e., "Res-L4 + TR-L3" and "Res-L4 + TR-L3W"). This is because the TR branch with stronger feature extraction ability was weakened, which resulted in much redundant information in the last-level feature maps of the backbone and made the training process of the proposed model more difficult. Moreover, when both the depth and the width of the backbone were increased (i.e., "Res-L4 + TR-L4" and "Res-L5 + TR-L4"), not only the number of parameters increased dramatically but also the performance did not improve, which could also be due to the expansion of the proportion of a simple CNN branch. To sum up, in view of the original intention to achieve the best balance between the model's performance and overhead, the proposed hybrid backbone configuration was set to "Res-L3 + TR-L3."

2) Discussion on the Pooling Scales of the TL-FFMs: Our TL-FFMs aim to fuse high-level context-rich features with

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TABLE III

ABLATION EXPERIMENTAL RESULTS OF THE PROPOSED HSSENet ON THE LEVIR, WHU, AND DSIFN TEST SETS. SER. AND PAR. MEAN THAT THE CNN AND TR BRANCHES WERE USED IN SERIES AND PARALLEL, RESPECTIVELY

Method	Module						Params (M)	FLOPs (G)	LEVIR		WHU		DSIFN	
	CNN	TR	Ser.	Par.	STEM	TL-FFM			F1	IoU	F1	IoU	F1	IoU
Base-L5	✓	-	-	-	-	-	11.85	25.97	89.84	81.61	89.68	81.29	67.05	50.43
Proposed 1	✓	✓	✓	-	-	-	8.55	16.54	90.21	82.16	90.04	81.88	85.07	74.04
Proposed 2	✓	✓	-	✓	-	-	**8.42**	**15.32**	90.91	83.34	90.24	82.21	86.70	76.53
Proposed 3	✓	✓	-	✓	✓	-	10.48	16.23	91.04	83.56	**90.65**	**82.90**	**89.69**	**81.31**
Proposed 4	✓	✓	-	✓	-	✓	**8.61**	**16.23**	**91.43**	**84.22**	90.55	82.73	88.13	78.78
HSSENet	✓	✓	-	✓	✓	✓	**10.95**	**17.14**	**91.48**	**84.29**	**91.55**	**84.42**	**91.17**	**83.77**

TABLE IV

ABLATION EXPERIMENTAL RESULTS OF THE PROPOSED HYBRID BACKBONE WITH DIFFERENT DEPTHS AND WIDTHS. NOTE THAT THE SA.N/CA.N OF THE STEM WAS SET TO 1/1

Model	B.N / B.C	Params /M	FLOPs /G	LEVIR		WHU		DSIFN	
				F1	IoU	F1	IoU	F1	IoU
Res-L3 + TR-L3	**4 / [64,64,128,256]**	**10.17**	**16.73**	**91.37**	**84.12**	**91.18**	**83.79**	**90.77**	**83.09**
Res-L3 + TR-L3W	4 / [64,128,256,512]	37.12	41.95	**91.49**	**84.32**	91.29	83.97	**92.21**	**85.55**
Res-L4 + TR-L3	4 / [64,64,128,256]	**14.03**	21.94	91.34	84.05	92.18	85.49	88.45	79.30
Base-L4 + TR-L3W	4 / [64,128,256,512]	45.12	49.27	**91.45**	**84.24**	**92.24**	**85.60**	90.47	82.60
Base-L4 + TR-L4	4 / [64,128,256,512]	40.64	**20.09**	90.98	83.45	91.61	84.51	89.70	81.32
Base-L5 + TR-L4	4 / [64,128,256,512]	53.75	25.00	90.95	83.41	**92.33**	**85.75**	89.52	81.04

TABLE V

ABLATION STUDY ON THE POOLING SCALES OF OUR TL-FFMs. NOTE THAT THE SA.N/CA.N OF STEM WAS SET TO 1/1

Pooling scales				LEVIR		WHU		DSIFN	
1	2	4	8	F1	IoU	F1	IoU	F1	IoU
✓				91.18	83.79	**90.70**	**82.99**	90.74	83.06
	✓			91.22	83.85	90.21	82.17	90.74	83.04
		✓		91.27	83.94	89.98	81.79	90.62	82.85
			✓	90.91	83.33	90.34	82.37	90.36	82.41
✓	✓			91.33	84.04	90.61	82.83	90.10	81.94
✓	✓	✓		**91.33**	**84.04**	90.63	82.87	**91.00**	**83.49**
✓	✓	✓	✓	**91.37**	**84.12**	**91.18**	**83.79**	**90.77**	**83.09**

low-level details-rich features in the gradual decoding process efficiently. They adopt the pooling results of the high-level features at different scales to guide the selection of low- and high-level features, so a critical problem is to determine appropriate parallel pooling scales. As shown in Table V, there were no great performance improvements by using different single pooling scales. However, when multiple pooling scales were used in parallel, consistent and stable performance on the three test sets was achieved, particularly when all four pooling scales were used. This could be because aggregating multilevel pooling information could provide different levels of local information and global information. Therefore, the edge details of the changing target can be refined while maintaining the intrinsic compactness of the overall structure in the gradual decoding process. Finally, the pooling scales of the proposed TL-FFMs were set to [1, 2, 4, 8].

3) Discussion on the Depth of the STEM: The depth of our STEM is a key hyperparameter, so a series of ablation experiments was conducted to examine the effects of the SA.N and CA.N in the STEM. As shown in Table VI, increasing the depth of the STI part provided certain benefits compared to the depth increase of the HLFE part. This indicated that the STEM required more TR layers to effectively encode the spatiotemporal dependencies between the bitemporal features. However, the HLFE in the STEM was efficient, with only one layer that could effectively enhance the representations of difference features. In general, the SA.N and CA.N were set to 6/1 and 2/1, respectively, to achieve superior performance. Still, when the SA.N increased from two to six, the Params increased by 3.16 M, with only 0.05 and 0.33 points of the F1 improvement on the LEVIR and WHU datasets, respectively. In addition, the performance on the complex DSIFN dataset was slightly degraded, which was because the two-layer self-attention was sufficient to establish the spatiotemporal relationships between bitemporal features. In conclusion, the values of SA.N and CA.N of the proposed STEM were set to 2 and 1, respectively.

4) Discussion on the Multiloss Function: Additional ablation experiments on the tradeoff parameters of the multiloss function (21) were conducted. It should be noted that the smaller the subscript of a parameter was, the farther it was from the final output of the proposed model. As shown in Table VII, the resultant metrics indicated that it was better to use multilevel predictions to guide model training than a

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TABLE VI
ABLATION EXPERIMENTAL RESULTS ON THE TR DEPTH OF OUR STEM

TR depth		Params /M	FLOPs /G	LEVIR		WHU		DSIFN	
SA.N	CA.N			F1	IoU	F1	IoU	F1	IoU
1	1	**10.17**	**16.73**	91.37	84.12	91.18	83.79	90.77	83.09
1	2	11.80	17.12	91.43	84.21	90.58	82.78	90.56	82.74
1	4	14.96	17.88	91.31	84.01	91.01	83.50	90.92	83.39
1	6	18.16	18.65	91.36	84.10	90.78	83.11	**91.12**	**83.69**
2	**1**	**10.95**	**17.14**	**91.48**	**84.29**	**91.55**	**84.42**	**91.17**	**83.77**
4	1	12.53	17.94	**91.51**	**84.35**	91.55	84.41	90.18	82.11
6	1	14.11	18.75	**91.53**	**84.38**	**91.88**	**84.98**	90.13	82.03

TABLE VII
ABLATION EXPERIMENTAL RESULTS ON MULTILOSS FUNCTION

Tradeoff parameters				LEVIR		WHU		DSIFN	
λ_1	λ_2	λ_3	λ_4	F1	IoU	F1	IoU	F1	IoU
1	**1**	**1**	**1**	**91.48**	**84.29**	**91.55**	**84.42**	**91.17**	**83.77**
0.7	0.8	0.9	1	**91.47**	**84.28**	**91.35**	**84.07**	**91.10**	**83.66**
0.4	0.6	0.8	1	91.45	84.24	91.23	83.87	90.91	83.33
0.1	0.4	0.7	1	91.40	84.17	91.16	83.76	91.03	83.54
0	0	0	1	90.98	83.45	90.73	83.04	90.77	83.10

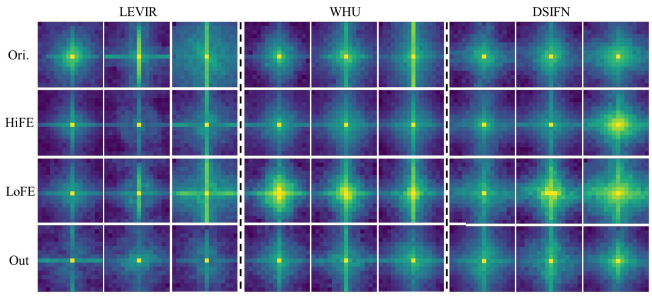


Fig. 12. Frequency magnitude visualization results (16×16) of three random output channels of four different parts in the HLFE. Ori. means original difference feature map $\mathbf{D}_4$; HiFE and LoFE denote high- and low-frequencies enhancement branches, respectively; and Out is the final output of the HLFE. The closer the pixel is to the center, the lower the frequency is, and the lighter the color is, the larger the magnitude is. Noted that the magnitude was averaged over 20 samples.

single-level prediction. Moreover, when multiloss functions with different tradeoff parameters were used, the model performance changed slightly. However, the smaller the interval between four-level prediction weights was, the better the model detection performance was. This could be because the difference feature enhancement and refinement in the deep layers also needed strong guidance from the real label. Consequently, $\lambda = [\lambda_1, \lambda_2, \lambda_3, \lambda_4]$ was set to [1, 1, 1, 1].

F. Spectrum Analysis of HLFE in STEM

The frequency magnitude of the output feature maps of different parts in the HLFE is presented in Fig. 12 to illustrate the effectiveness of the HLFE in the STEM. As shown in Fig. 12, the original difference feature ($\mathbf{D}_4$) derived from the subtraction of bitemporal features contained a certain degree of high-frequency noise interference. After the HiFE processing, the high-frequency noise was suppressed, which was reflected in the spectrum diagram as the pixel far away from the center darkens. Further, after the LoFE processing, the integrity and smoothness of the internal structure of the changed targets were enhanced, and thus, the low-frequency of features was enhanced. It was reflected in the brightening of the middle region on the spectrum diagram. To make full use of the advantages of the two branches, we used the sum of them as the output of HLFE, which can not only filter the noise to a certain extent, but also enhance the internal compactness of the changed target. Note that the role of the HLFE (or the STEM) in maintaining the compact structure of changed targets and suppressing high-frequency noise and pseudo-changes can be clearly seen in the spatial domain visualization in Fig. 13(STEM).

G. Network Visualization Analysis

To provide a better visual explanation of the prediction results of each stage of our HSSENet, we applied the gradient-weighted class activation mapping (Grad-CAM) [71], [72] to visualize the output of each module, and further verified their effectiveness. The Grad-CAM is a visualization method that uses the gradients to weigh the feature maps and evaluates the importance of spatial positions in the target layer for the predicted CD maps. The visualization results of our HSSENet in different stages can be seen in Fig. 13.

First, the four representative feature maps obtained from the last-level outputs of our hybrid backbone are presented in the middle column of Fig. 13(a) and (b). To illustrate the strong feature extraction capability further, the input images with the Grad-CAM mask of the backbone are shown in the last column in Fig. 13(a) and (b). The results indicated that our hybrid backbone focused on objects of interest related to the CD task, and clearly distinguished the interested (e.g., the buildings on the LEVIR and WHU datasets and the buildings, croplands, and roads on the DSIFN dataset) and irrelevant components from the bitemporal inputs due to the efficient combination of the CNN and TR. Subsequently, the Grad-CAM was used to visualize the output at different stages of our HSSENet. As shown in Fig. 13(Diff), the last-level feature maps output by the encoder were ambiguous. They contained many irrelevant changes and high-frequency burrs,

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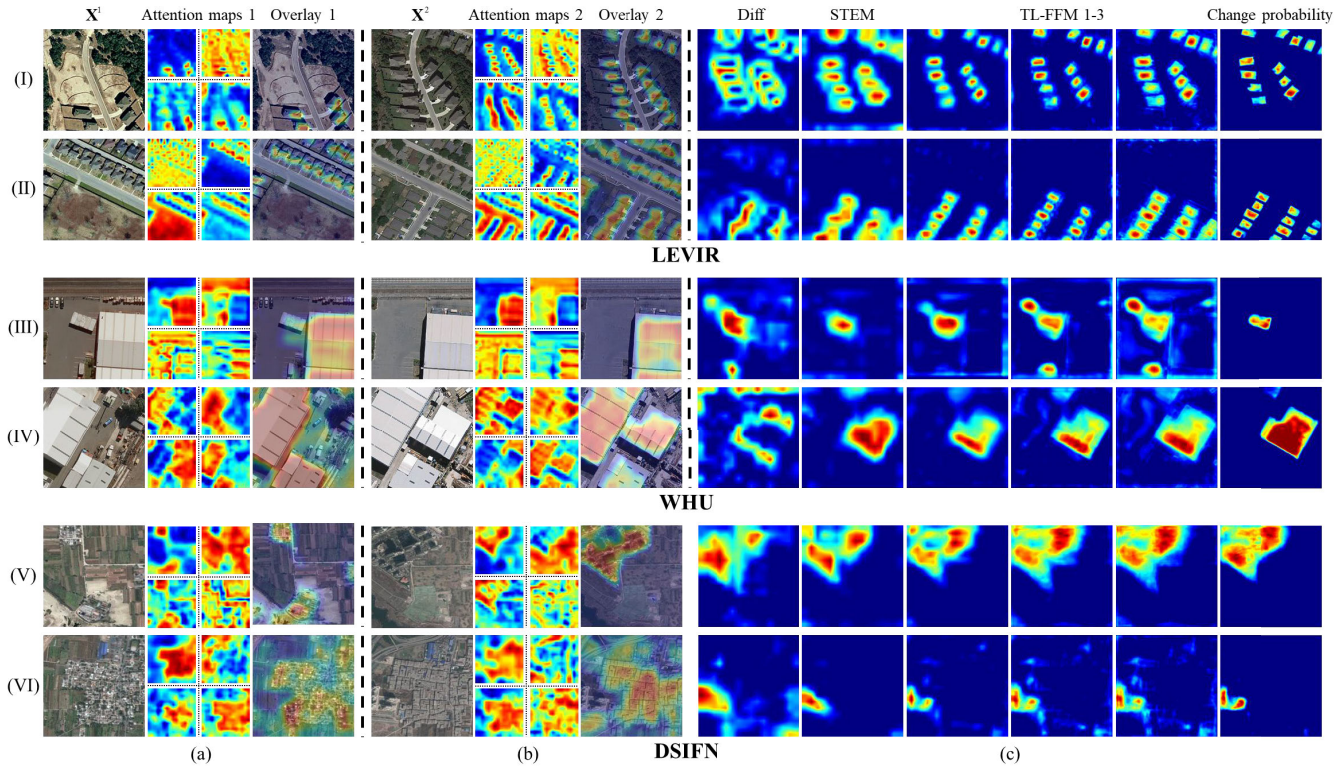


Fig. 13. Grad-CAM visualization results on the LEVIR/WHU/DSIFN datasets: (a) $\mathbf{X}^1$, four selected representative feature maps of $\mathbf{X}^1$, $\mathbf{X}^1$ with the Grad-CAM mask of the Siamese backbone[1]; (b) $\mathbf{X}^2$, four selected representative feature maps of $\mathbf{X}^2$, $\mathbf{X}^2$ with the Grad-CAM mask of the Siamese backbone[2]; and (c) visualization results of each module in decoding, where red and blue indicate higher and lower attention values, respectively.

and had low tightness of changed objects. After the STEM, as shown in Fig. 13(STEM), the internal compactness of the changed targets was improved due to the low-frequency enhancement by the STEM, and the pseudo-changes and high-frequency burrs were significantly suppressed due to the high-frequency enhancement by the STEM. Then, the difference features generated by the STEM were fed to the three TL-FFMs to refine boundary details. The results of the gradual refinement of changes can be seen in Fig. 13(TL-FFM 1-3). For instance, as shown in Fig. 13(I)–(IV), the originally elliptic changed buildings gradually became regular square targets, and the boundary contour became more detailed, which was much closer to the label. The results also demonstrated that fusing multilevel information could be critical for refining the changed objects of interest. Interestingly, as presented in Fig. 13(III), our HSSENet highlighted the missing vehicles although no relevant supervisory information was provided during the model training. This indicated that the proposed HSSENet could learn the implicit change information. As shown in Fig. 13(VI), our model could retrieve small changed objects in the stepwise decoding process. Furthermore, due to the introduction of HR features with abundant details in the decoding process, the difference features contained a certain degree of high-frequency noise. However, after using the proposed specially designed CD head, high-frequency interference was successfully filtered out as shown in the last column of Fig. 13(c).

Consequently, the visualization results of each module have proven that the proposed hybrid Siamese backbone, STEM, and TL-FFMs are all indispensable parts of the high-precision CD. In addition, all of the proposed modules showed excellent performances on the three public datasets, which indicates their strong robustness.

V. Conclusion

In this article, the HSSENet is proposed to address the main problems of the existing methods for RS image CD by introducing three main modules. First, the CNN and TR are combined for feature extraction to obtain hierarchical discriminative features from the bitemporal images with relatively low model overhead. In addition, the proposed STEM can effectively eliminate the pseudo-changes due to the context modeling in spatiotemporal dimensions and achieve feature enhancement in HiLo frequencies. Finally, the three TL-FFMs can not only gradually refine the boundary details of changed objects but also maintain the compact internal structure. The performance of the proposed method and the effect of each of our modules are evaluated by extensive experiments. The results show that compared with the SOTA methods, our HSSENet can achieve better performance. The F1-score of the proposed model even exceeds the second-best model by 0.54, 1.58, and 3.04 points on the LEVIR, WHU, and DSIFN datasets. More importantly, the proposed HSSENet can achieve a better tradeoff between accuracy, parameter number, and FLOPs than the SOTA methods.

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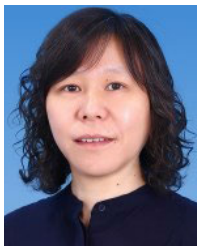
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